PORTFOLIO PROJECT · DATA ANALYTICS

# RETAIL SALES ANALYSIS

Python, MySQL, and Power BI on Superstore Data

10,194 sales lines · 2023 to 2026 transaction records

CORE TAKEAWAY

## SALES VOLUME TELLS ONLY HALF THE STORY.

Tracking profit alongside sales reveals where revenue generates return and where margins leak.



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THE CORE CHALLENGE

# HIGH SALES ALONE DON'T GUARANTEE STRONG PROFIT

A product line or region can generate substantial top-line revenue while contributing little to bottom-line profit.

Decision-makers need visibility into where revenue turns into real return across products, customer segments, regions, and time.

ANALYSIS OBJECTIVE

## Evaluate sales and profit jointly

Pinpoint exact categories and territories that warrant investigation, diagnose discount exposure, and improve overall profitability.




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# ONE DATASET, THREE TOOLS, ONE WORKFLOW

Sample Superstore dataset · 10,194 sales-line records · 21 columns · January 3, 2023 to December 30, 2026

## 01

### Python

**AUDIT & VALIDATION**

Validate the data and analyze sales and profit.

## 02

### MySQL (XAMPP)

**RELATIONAL STORAGE**

Store the dataset for SQL querying and analysis.

## 03

### Power BI

**INTERACTIVE DASHBOARD**

Present the analysis in an interactive dashboard.

Fields: order and ship dates, customer, product, segment, category, sub-category, region, sales, quantity, discount, profit. Not included: age, gender, subscription, review, or sales-channel fields.

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# PYTHON CHECKS FOUND A CLEAN DATASET

DATA HYGIENE

**0**

## Missing values or duplicate rows

No fully duplicated rows; Row ID is unique.

| Check | Values | Finding |
|---|---|---|
| **Sales per line** | Median $53.91; mean $228.23 | A few high-value lines pull the average up |
| **Profit per line** | -$6,599.98 to $8,399.98 | Records to investigate, not confirmed errors |
| **Discount** | Mean 15.54%; median 20%; maximum 80% | — |
| **Parsed types** | Dates as dates; Sales, Profit, Discount as decimals; Quantity as whole numbers | |

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# $2.33M IN SALES BECAME $292K IN PROFIT

**GROSS SALES**

Total recorded revenue

**$2.33M**

$2,326,534.35 exact

**NET PROFIT**

Total bottom-line profit

**$292K**

$292,296.81 exact

**OVERALL MARGIN**

Total profit ÷ total sales

**12.6%**

12.56% portfolio return

**VOLUME SOLD**

Total transaction units

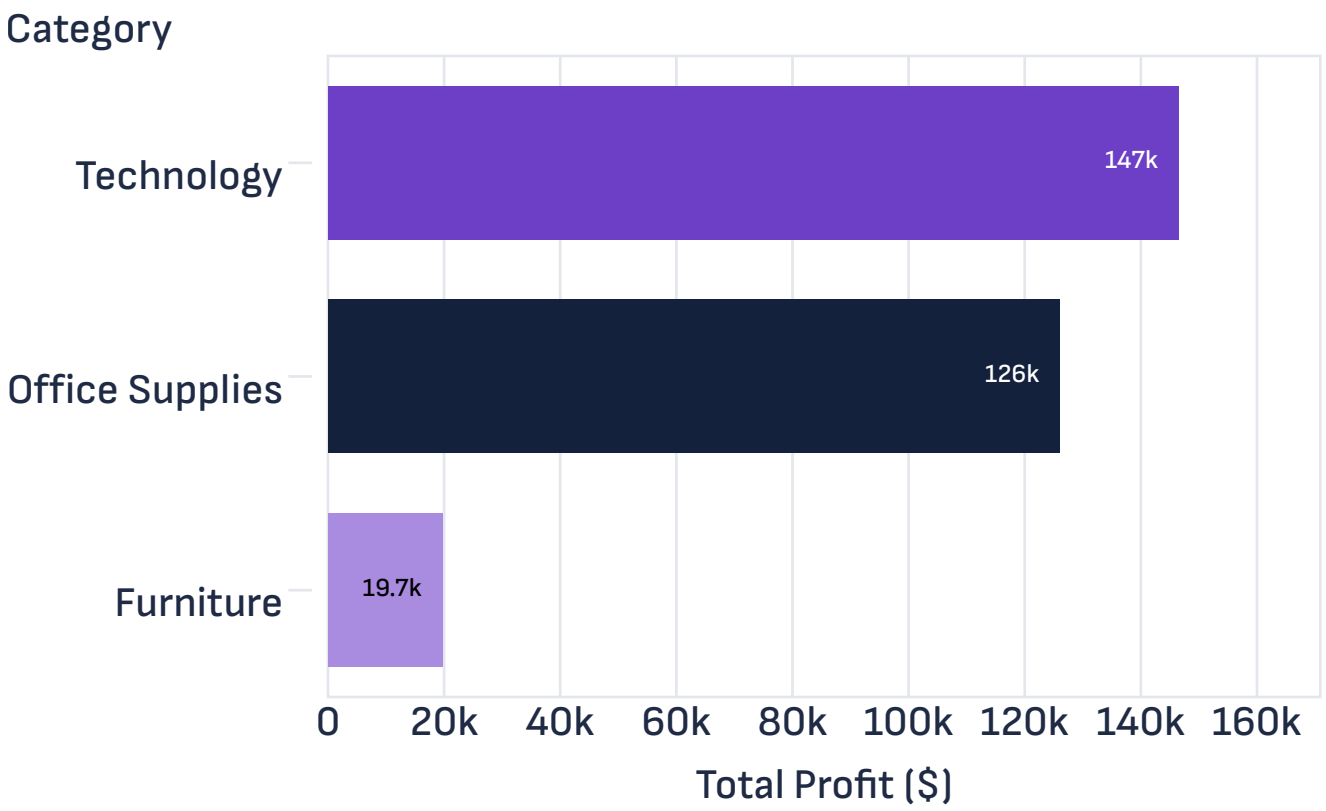
**38.7K**

38,654 total units

**Executive takeaway:** Across 10,194 order lines, the store converts roughly 12.6 cents of every sales dollar into profit. The following slides analyze how this conversion varies radically across categories and geographic regions.

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# TECHNOLOGY AND OFFICE SUPPLIES DRIVE NEARLY ALL PROFIT




Total profit dollars across all lines. These reflect total profit, not category profit margins.

CATEGORY PROFIT SHARE

## Tech accounts for 50.1% of profit

**Technology:** $146,543.38 total profit, averaging $78.58 per sales line.

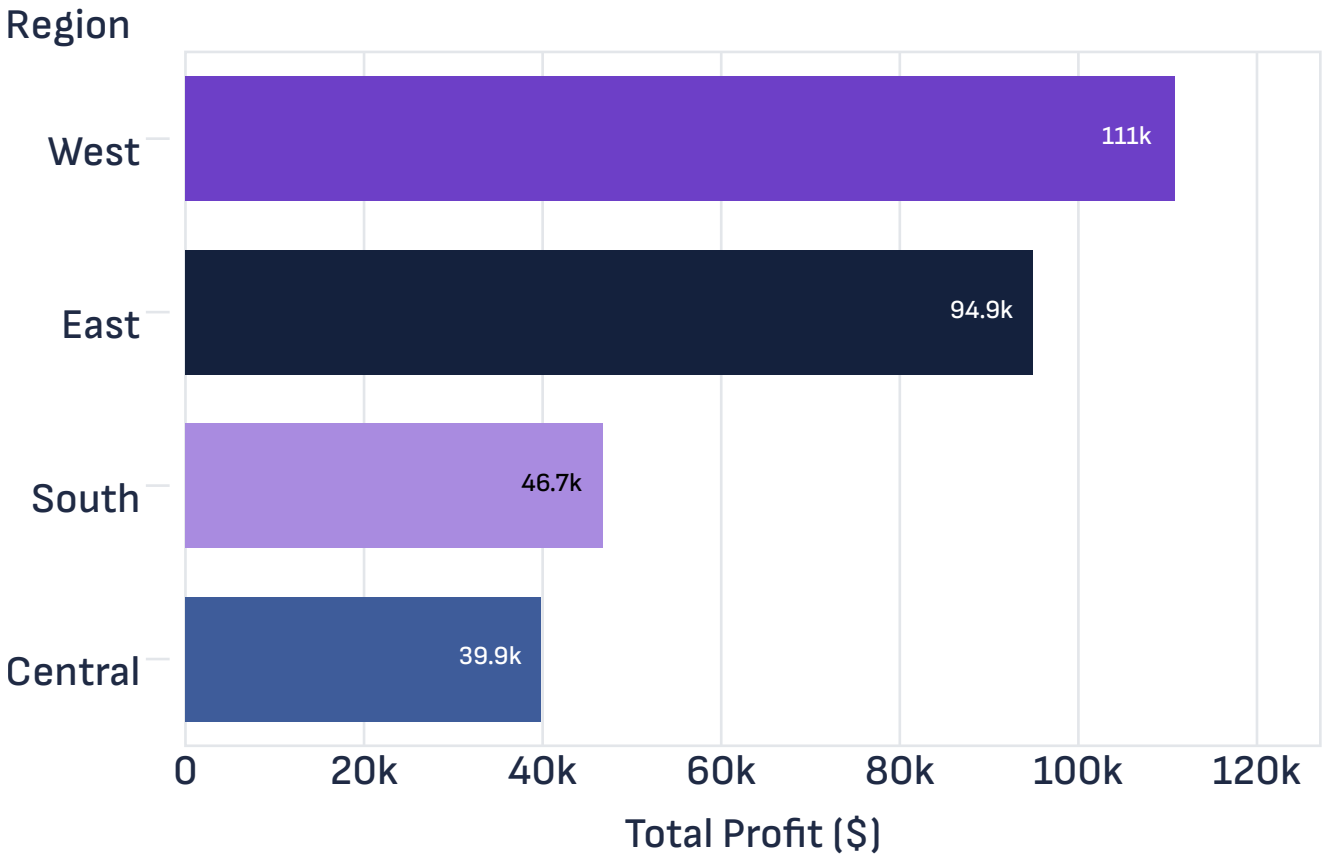
**Office Supplies:** $126,023.44 total (43.1%), averaging $20.57 per line.

**Furniture:** $19,730.00 total (6.7%), averaging just $8.96 per line.

Together, Tech and Office Supplies generate 93.3% of total enterprise profit.

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# WEST AND EAST GENERATE ABOUT 70% OF PROFIT




Total profit dollars by geographic region. These figures are totals and per-line averages, not profit margins.

REGIONAL CONTRIBUTION

## West & East total 70.4%

**West:** $110,798.82 total · $34.06 avg profit/line

**East:** $94,883.26 total · $31.78 avg profit/line

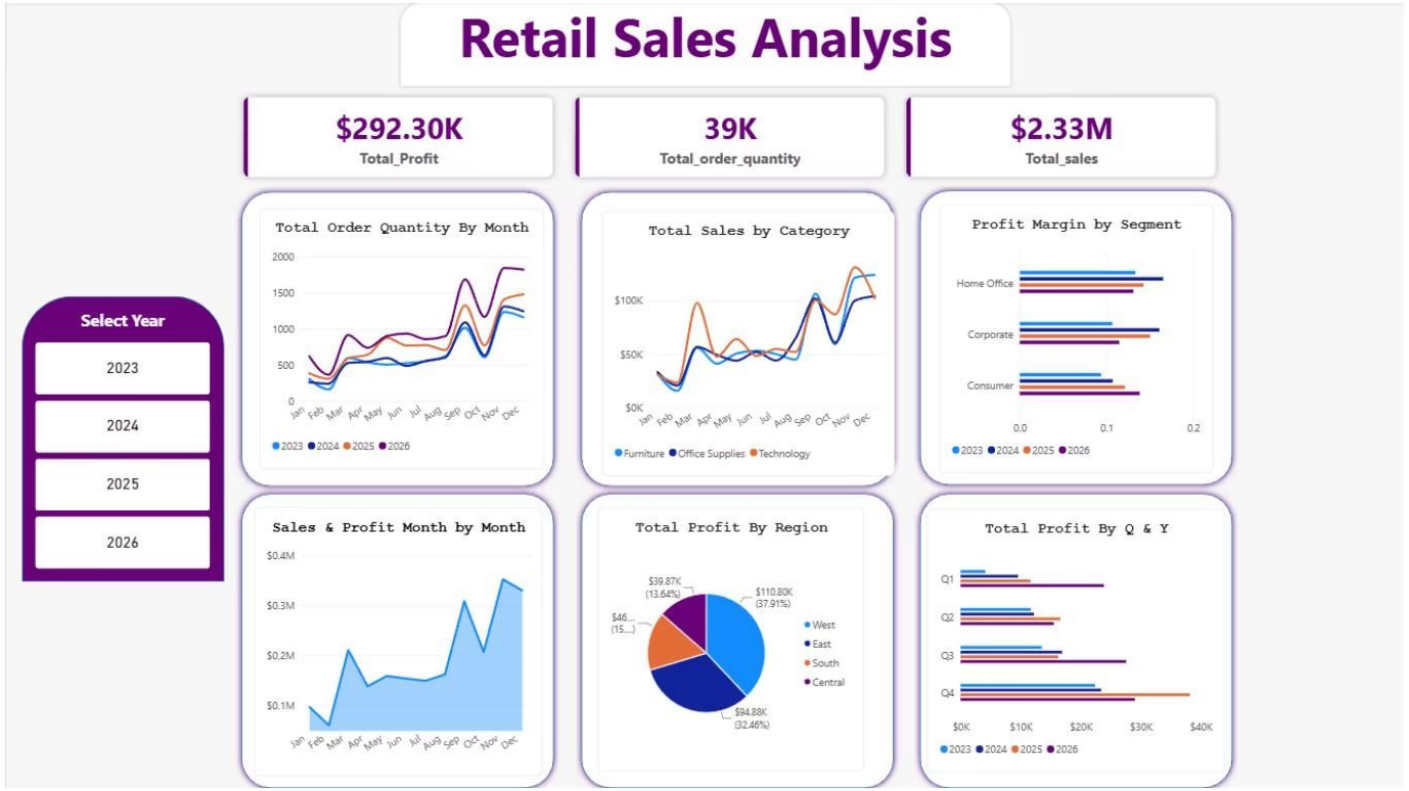
**South:** $46,749.43 total · $28.86 avg profit/line

**Central:** $39,865.31 total · $17.07 avg profit/line

Central ranks lowest across both total profit dollars and average return per order line.

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# THE POWER BI DASHBOARD LETS YOU EXPLORE THE SAME QUESTIONS




INTERACTIVE FEATURES

## Multi-dimensional views

**KPI Cards:** Dynamic totals for sales, profit, and quantity with year slicers.

**Visual breakdowns:** Monthly quantity, category sales, segment margin, and regional profit distribution.

**DASHBOARD REFINEMENT**

The chart titled "Sales & Profit Month by Month" currently displays only sales. Add the profit series or rename it "Total Sales by Month".

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# WHERE TO LOOK NEXT

Actionable investigative paths derived from descriptive patterns

### Technology sub-categories

Isolate which specific product lines (such as Phones, Copiers, or Accessories) drive the 50.1% profit share and protect their margin health.

### Office Supplies volume & discount

Evaluate high-volume sub-categories with lower profit per line to determine if promotional discounting is eroding potential margin.

### Furniture loss-making lines

Audit individual loss-making order lines and discount depth across Tables and Bookcases before altering product assortment.

### Central region product mix

Compare Central's average line return ($17.07) against West ($34.06) and East ($31.78) to identify whether product mix or discount policies differ.

**Analytic caveat:** These are descriptive findings highlighting operational priorities; they identify where performance differs, not the root causal drivers.

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